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Studierichting: Informatica
Oefeningenreeks 1C nr. 12.1

Los $\frac{-1+i}{\sqrt{2}}$ op

$$z_1 = -1 + i = r_1 \operatorname{cis}(\alpha)$$

$$\text{en } z_2 = r_2 \operatorname{cis}(\beta)$$

Moduli:

$$r_1 = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$r_2 = \sqrt{2}$$

Argumenten:

$$\alpha = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{1}{-1}\right) = \tan^{-1}(-1) = \frac{3\pi}{4}$$

$$\beta = \tan^{-1}\left(\frac{0}{\sqrt{2}}\right)$$

Goniometrische vorm:

$$z_1 = \sqrt{2} \operatorname{cis}\left(\frac{3\pi}{4}\right)$$

$$z_2 = \sqrt{2} \operatorname{cis}(0)$$

Deling:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\alpha - \beta) = \operatorname{cis}\left(\frac{3\pi}{4}\right)$$

References